National University of Singapore

MA1522 - Linear Algebra for Computing

Final Exam

(AY2024/2025 Semester 1)

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS

1. This is a NON-SECURE BLOCK INTERNET assessment. You are allowed to refer to any soft copy or hard copy notes. All soft copy notes are to be access through the device you are using for the exam.
2. This assessment contains 2 sections for a total of **80 marks**. You are required to answer ALL questions.
3. There are **15** questions in Section A, with **3 marks** per question. For each question, choose all the correct options.
4. Submit your answers for Section A in Examplify.
5. Section B consistent of **4** (**FOUR**) questions, and a total of **35 marks**.
6. The questions for section B can be found in an attachment in question 16 of Examplify.
7. Write your answers for section B in the answer booklet provided.
8. Write your student number (e.g. A1234567Z) on the cover page of the answer booklet, in the space provided. **Do not write your name**.
9. **Begin each question on a new page** and indicate the question number clearly at the top of each page.
10. You are allowed to use MATLAB and any offline calculator. However, you should lay out systemically the various steps in the calculations for section B.
11. Hand in all the answer booklet at the end of the examination.
12. You are not allowed exit Examplify without informing the examiner and obtaining the exit password. Type in the exit password in the blanks provided in question 16 in Examplify.

Section A

1. (3 points) Let $\mathbf{A}$ be a 3×5 matrix. Given that a non-homogeneous system $\mathbf{Ax} = \mathbf{b}$ has the general solution

$$\begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{pmatrix} = \begin{pmatrix} 1+s-t \\ s+t \\ s \\ t+2 \\ t \end{pmatrix}.$$

Which of the following statements are correct?

(a) $\begin{pmatrix} 2 \\ 1 \\ 1 \\ 2 \\ 0 \end{pmatrix}$ is a solution to $\mathbf{Ax} = \mathbf{b}$.

(b) $\begin{pmatrix} 0 \\ 2 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ is a solution to the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

(c) $\begin{pmatrix} 1 \\ 2 \\ 1 \\ 3 \\ 1 \end{pmatrix}$ is a solution to the homogeneous system $\mathbf{Ax} = \mathbf{0}$.

(d) $\begin{pmatrix} -1 \\ 3 \\ 1 \\ 2 \\ 2 \end{pmatrix}$ is a solution to $\mathbf{Ax} = \mathbf{b}$.

2. (3 points) Let

$$\mathbf{A} = \mathbf{L} \begin{pmatrix} 2 & 1 & 1 & -2 \\ 0 & 0 & 3 & 1 \\ 0 & 0 & 0 & 5 \end{pmatrix}$$

be an LU-factorization of a 3×4 matrix $\mathbf{A}$. Given that

$$\mathbf{A}\begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ -6 \end{pmatrix}, \quad \mathbf{A}\begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \\ 1 \end{pmatrix}, \quad \mathbf{A}\begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}.$$

Which of the following are correct?

Question 2 continues...

(a) $\mathbf{A}\begin{pmatrix}0\\1\\0\\0\end{pmatrix} = \begin{pmatrix}1\\2\\-3\end{pmatrix}$

(b) $\mathbf{A}\begin{pmatrix}1\\0\\2\\0\end{pmatrix} = \begin{pmatrix}4\\10\\-4\end{pmatrix}$

(c) $\mathbf{A}\begin{pmatrix}0\\0\\1\\1\end{pmatrix} = \begin{pmatrix}1\\2\\1\end{pmatrix}$

(d) $\mathbf{A}\begin{pmatrix}1\\0\\0\\2\end{pmatrix} = \begin{pmatrix}2\\4\\-6\end{pmatrix}$

3. (3 points) Let

$$\mathbf{A} = \mathbf{L}\begin{pmatrix}2 & 1 & 1 & -2\\0 & 0 & 3 & 1\\0 & 0 & 0 & 5\end{pmatrix}$$

be an LU-factorization of a 3×4 matrix $\mathbf{A}$. Write

$$\mathbf{A} = \begin{pmatrix}\mathbf{a}_1 & \mathbf{a}_2 & \mathbf{a}_3 & \mathbf{a}_4\end{pmatrix},$$

where $\mathbf{a}_i$ is the i-th column of $\mathbf{A}$, for $i = 1, 2, 3, 4$. Which of the following sets are bases for the column space of $\mathbf{A}$?

(a) $\{\mathbf{a}_1, \mathbf{a}_3, \mathbf{a}_4\}$

(b) $\{\mathbf{a}_2, \mathbf{a}_3, \mathbf{a}_4\}$

(c) $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_4\}$

(d) $\{\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3\}$

4. (3 points) Given that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is a basis for a subspace V of $\mathbb{R}^n$. Which of the following sets are bases for V?

(a) $\{2\mathbf{u}_1 + \mathbf{u}_3, \mathbf{u}_1 + \mathbf{u}_2, \mathbf{u}_1 + 2\mathbf{u}_2\}$

(b) $\{\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3, 2\mathbf{u}_1 - \mathbf{u}_3, \mathbf{u}_1 - 2\mathbf{u}_2 + \mathbf{u}_3\}$

(c) $\{\mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3, \mathbf{u}_1 - \mathbf{u}_2, \mathbf{u}_1 + \mathbf{u}_2 + \mathbf{u}_3\}$

(d) $\{\mathbf{u}_1 - \mathbf{u}_2 + \mathbf{u}_3, \mathbf{u}_1 + \mathbf{u}_2, 2\mathbf{u}_1 + \mathbf{u}_3\}$

5. (3 points) Suppose

$$V = \left\{ \begin{pmatrix}x_1\\x_2\\x_3\\x_4\end{pmatrix} \;\middle|\; \left\{ \begin{array}{rcrcrcrcl} x_1 & & & - & 2x_3 & - & 2x_4 & = & 0\\ 3x_1 & + & 4x_2 & + & 4x_3 & + & 4x_4 & = & 0\\ 2x_1 & + & 2x_2 & + & x_3 & + & x_4 & = & 0 \end{array} \right. \right\}.$$

Question 5 continues...

Which of the following sets are bases for V?

(a) $\left\{ \begin{pmatrix} 4 \\ -5 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ -5 \\ 0 \\ 2 \end{pmatrix} \right\}$

(b) $\left\{ \begin{pmatrix} 0 \\ 0 \\ -2 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -5 \\ 2 \\ 0 \end{pmatrix} \right\}$

(c) $\left\{ \begin{pmatrix} 4 \\ 5 \\ 0 \\ -2 \end{pmatrix} \right\}$

(d) $\left\{ \begin{pmatrix} 0 \\ 0 \\ -2 \\ 2 \end{pmatrix} \right\}$

(e) $\left\{ \begin{pmatrix} 4 \\ 5 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 4 \\ -5 \\ 2 \\ 0 \end{pmatrix} \right\}$

6. (3 points) Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthogonal (not necessary orthonormal) basis for a subspace V of $\mathbb{R}^n$. Then for any $\mathbf{x}, \mathbf{y}$ in V,

   (a) $[\mathbf{x}]_S + [\mathbf{y}]_S = [\mathbf{x} + \mathbf{y}]_S$.

   (b) $\alpha[\mathbf{x}]_S = [\alpha\mathbf{x}]_S$ for any real number α.

   (c) $[\mathbf{x}]_S \cdot [\mathbf{y}]_S = \mathbf{x} \cdot \mathbf{y}$.

   (d) $\|[\mathbf{x}]_S\| = \|\mathbf{x}\|$.

7. (3 points) Let $S = \left\{ \begin{pmatrix} 0 \\ -1 \\ 2 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 2 \\ 2 \\ 0 \\ -1 \end{pmatrix} \right\}$ and $T = \left\{ \begin{pmatrix} 2 \\ 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 2 \\ -1 \end{pmatrix} \right\}$.

   Which of the following statements are true?

   (a) $\begin{pmatrix} -2 & 1 & 2 \\ 4 & -2 & -2 \\ -1 & 1 & 1 \end{pmatrix}$ is the transition matrix from T to S.

   (b) $\begin{pmatrix} 0 & 1/2 & 1 \\ -1 & 0 & 2 \\ 1 & 1/2 & 0 \end{pmatrix}$ is the transition matrix from S to T.

Question 7 continues...

(c) $\begin{pmatrix} -2 & 1 & 2 \\ 4 & -2 & -2 \\ -1 & 1 & 1 \end{pmatrix}$ is the transition matrix from S to T.

(d) $\begin{pmatrix} 0 & 1/2 & 1 \\ -1 & 0 & 2 \\ 1 & 1/2 & 0 \end{pmatrix}$ is the transition matrix from T to S.

(e) S is not a basis for any subspace.

(f) T is not a basis for any subspace

8. (3 points) Suppose

$$\mathbf{A} = \begin{pmatrix} 2 & 2 & 1 & 3 \\ 0 & 1 & 1 & 0 \\ 2 & 2 & 1 & -1 \end{pmatrix}.$$

Which of the following matrices have the same row space as $\mathbf{A}$?

(a) $\begin{pmatrix} 2 & 5 & 4 & -1 \\ 6 & 6 & 3 & 5 \\ 8 & 10 & 6 & 8 \end{pmatrix}$

(b) $\begin{pmatrix} 2 & 2 & 1 & 3 \\ 8 & 7 & 3 & 4 \\ 2 & 5 & 4 & 7 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 3 & 3 & -4 \\ 12 & 15 & 9 & 6 \\ 8 & 7 & 3 & 8 \end{pmatrix}$

(d) $\begin{pmatrix} 6 & 7 & 4 & -3 \\ 2 & 13 & 7 & 6 \\ 4 & 4 & 2 & 2 \end{pmatrix}$

9. (3 points) Two vectors $\mathbf{u}, \mathbf{v}$ are said to be collinear if there is a constant α such that $\mathbf{u} = \alpha\mathbf{v}$, or there is a constant β such that $\mathbf{v} = \beta\mathbf{u}$.

Suppose $\mathbf{A}$ is a 4×3 matrix and $\mathbf{B}$ is a 3×3 matrix such that there are non-collinear vectors $\mathbf{u}, \mathbf{v}$ with $\mathbf{Au} = \mathbf{0}$ and $\mathbf{Av} = \mathbf{0}$. Which of the following statements are true?

(a) If $\mathbf{b}$ and $\mathbf{c}$ are such that $\mathbf{ABx} = \mathbf{b}$ and $\mathbf{ABx} = \mathbf{c}$ are consistent, then $\mathbf{b}$ and $\mathbf{c}$ are collinear.

(b) We can find non-colinear vectors $\mathbf{w}_1, \mathbf{w}_2$ such that $\mathbf{ABw}_1 = \mathbf{0}$ and $\mathbf{ABw}_2 = \mathbf{0}$.

(c) We can find non-colinear vectors $\mathbf{b}$ and $\mathbf{c}$ are such that $\mathbf{ABx} = \mathbf{b}$ and $\mathbf{ABx} = \mathbf{c}$.

(d) If $\mathbf{w}_1, \mathbf{w}_2$ are such that $\mathbf{ABw}_1 = \mathbf{0}$ and $\mathbf{ABw}_2 = \mathbf{0}$, then $\mathbf{w}_1$ and $\mathbf{w}_2$ are collinear.

10. (3 points) Suppose

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{3} \\ 0 & 1/\sqrt{2} & -1/\sqrt{3} \\ 0 & 0 & 1/\sqrt{3} \end{pmatrix} \mathbf{R}$$

is a QR factorization of $\mathbf{A}$. Which of the following statements are true?

(a) The row space of $\mathbf{A}$ is $\mathbb{R}^3$.

(b) The projection of $\mathbf{w} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ onto the column space of $\mathbf{A}$ is $\begin{pmatrix} 1 \\ 4/3 \\ 2/3 \\ 1/3 \end{pmatrix}$.

(c) The least squares solution to $\mathbf{Ax} = \begin{pmatrix} 1 \\ 1 \\ 1 \\ 1 \end{pmatrix}$ is $\begin{pmatrix} 1 \\ 4/3 \\ 2/3 \\ 1/3 \end{pmatrix}$.

(d) $\mathbf{AA}^T$ is invertible.

11. (3 points) Let

$$\mathbf{A} = \begin{pmatrix} -1 & -1 & 0 & 1 \\ 1 & 1 & -1 & -1 \\ 2 & 0 & 2 & 1 \\ 2 & -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} a & 0 & 0 & 0 \\ 0 & a & 0 & 0 \\ 0 & 0 & a & 0 \\ 0 & 0 & 0 & b \end{pmatrix} \begin{pmatrix} -1 & -1 & 0 & 1 \\ 1 & 1 & -1 & -1 \\ 2 & 0 & 2 & 1 \\ 2 & -1 & 0 & 0 \end{pmatrix}^{-1}$$

for some constants a, b. Which of the are eigenvectors of $\mathbf{A}$?

(a) $\begin{pmatrix} 2 \\ -1 \\ -2 \\ 2 \end{pmatrix}$

(b) $\begin{pmatrix} 4 \\ -6 \\ 2 \\ 1 \end{pmatrix}$

(c) $\begin{pmatrix} 2 \\ -5 \\ 6 \\ 2 \end{pmatrix}$

(d) $\begin{pmatrix} 3 \\ -5 \\ 3 \\ 6 \end{pmatrix}$

Question 11 continues...

(e) $\begin{pmatrix} -4 \\ 3 \\ 4 \\ 4 \end{pmatrix}$

12. (3 points) Let $\mathbf{P} = \begin{pmatrix} \mathbf{v}_1 & \mathbf{v}_2 & \mathbf{v}_3 & \mathbf{v}_4 \end{pmatrix}$ be a 4×4 invertible matrix, where $\mathbf{v}_i$ is the i-th column of $\mathbf{P}$ for $i = 1, 2, 3, 4$. Given that

$$\mathbf{A} = \mathbf{P} \begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & -1 \\ 1 & 0 & 1 & 0 \\ 0 & -1 & 0 & 1 \end{pmatrix} \mathbf{P}^{-1}.$$

Which of the following are eigenvectors of $\mathbf{A}$?

(a) $\mathbf{v}_1 + \mathbf{v}_3$

(b) $\mathbf{v}_2 - \mathbf{v}_4$

(c) $\mathbf{v}_2 + \mathbf{v}_4$

(d) $\mathbf{v}_1 - \mathbf{v}_2$

(e) $\mathbf{v}_1 + \mathbf{v}_2$

(f) $\mathbf{v}_3 - \mathbf{v}_4$

13. (3 points) A certain island experiences only 3 types of weather in a day, sunny , cloudy, or rain. It is known that the weather on a following day depends on the whether in the previous day as such,

(i) if it was sunny yesterday, then there is a 30% chance it will remain sunny today, and 40% chance it will be cloudy today;

(ii) if it was cloudy yesterday, then there is a 40% chance it will remain cloudy today, and 40% chance it will be raining today;

(iii) if it was raining yesterday, then there is a 50% chance it will continue to rain today, and 20% chance it will be sunny today.

In the long run, which whether would the island experience most, 1st being the highest occurrence of the whether experienced.

(a) 1. raining 2. cloudy 3. sunny

(b) 1. cloudy 2. raining 3. sunny

(c) 1. sunny 2. cloudy 3. raining

(d) 1. raining 2. sunny 3. cloudy

(e) 1. cloudy 2. sunny 3. raining

14. (3 points) Let $\mathbf{A}$ be a 4×3 matrix such that $\mathbf{A}^T\mathbf{A} = \begin{pmatrix} 5 & -2 & 0 \\ -2 & 8 & 0 \\ 0 & 0 & 4 \end{pmatrix}$. Which of the following are the singular values of $\mathbf{A}$?

Question 14 continues...

(a) 2

(b) 3

(c) 4

(d) 9

(e) 0

15. (3 points) Let $S = \{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be a basis for $\mathbb{R}^3$ and $T : \mathbb{R}^3 \to \mathbb{R}^4$ be a linear transformation such that

$$T(\mathbf{u}_1) = \begin{pmatrix} 2 \\ 2 \\ 1 \\ 5 \end{pmatrix}, \quad T(\mathbf{u}_2) = \begin{pmatrix} 1 \\ -1 \\ 4 \\ 1 \end{pmatrix}, \quad T(\mathbf{u}_3) = \begin{pmatrix} 0 \\ 1 \\ -1 \\ -1 \end{pmatrix}.$$

Let $\mathbf{w}$ be a vector in $\mathbb{R}^3$ and $[\mathbf{w}]_S$ be its coordinates relative to basis S. Which of the following are correct?

(a) $T(\mathbf{w}) = \begin{pmatrix} 1 \\ 6 \\ -6 \\ 1 \end{pmatrix}$ if $[\mathbf{w}]_S = \begin{pmatrix} 1 \\ -1 \\ 3 \end{pmatrix}$

(b) $T(\mathbf{w}) = \begin{pmatrix} 1 \\ 1 \\ 2 \\ -1 \end{pmatrix}$ if $[\mathbf{w}]_S = \begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$

(c) $T(\mathbf{w}) = \begin{pmatrix} 3 \\ 2 \\ 5 \\ 7 \end{pmatrix}$ if $[\mathbf{w}]_S = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

(d) $T(\mathbf{w}) = \begin{pmatrix} 1 \\ 3 \\ 3 \\ 4 \end{pmatrix}$ if $[\mathbf{w}]_S = \begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}$

Section B

16. Given that $\mathbf{A} = \mathbf{L}\begin{pmatrix} -1 & 1 & 7 & -1 \\ 0 & 4 & 12 & 2 \\ 0 & 0 & 0 & -1 \end{pmatrix}$ is an LU factorization of a 3×4 matrix.

(a) (6 points) It is given that $\mathbf{A}\begin{pmatrix} 1 \\ -1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 14 \\ 17 \end{pmatrix}$ and $\mathbf{A}\begin{pmatrix} 0 \\ 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 7 \end{pmatrix}$. Find $\mathbf{A}$.

(b) (4 points) It is given that $\begin{pmatrix} -1 \\ 0 \\ 0 \\ 0 \end{pmatrix}$ is a least square solution to the system $\mathbf{Ax} = \begin{pmatrix} 1 \\ 1 \\ 2 \end{pmatrix}$.

Use your answer in (a) or otherwise, find all the least square solutions to the system.

17. Consider the following data

x	-1	-0.15	0	1	2.15	3	3.3
y	0	3	3	0	-3	0	3

(a) (2 points) Is it possible to find a degree 6 polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + a_6x^6$$

that passes through all the data points? You do not need to justify your answer.

(b) (2 points) Is it possible to find a degree 3 polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

that passes through all the data points? You do not need to justify your answer.

(c) (2 points) Find a degree 3 polynomial

$$p(x) = a_0 + a_1x + a_2x^2 + a_3x^3$$

that is the least squared approximation to the data. Give your answer to the nearest 3 significant figure, you do not need to show your working.

18. A 4×3 matrix $\mathbf{A}$ has SVD decomposition

$$\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T,$$

where $\mathbf{V} = \begin{pmatrix} 0 & 2/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & -1/\sqrt{6} & -1/\sqrt{3} \\ 1/\sqrt{2} & 1/\sqrt{6} & 1/\sqrt{3} \end{pmatrix}$, and such that the characteristic polynomial of $\mathbf{A}^T\mathbf{A}$ is $(x-3)(x-6)(x-10)$.

(a) (2 points) Find $\mathbf{\Sigma}$?

Question 18 continues...

(b) (5 points) It is given that

$$\frac{1}{2\sqrt{5}}\mathbf{A}\begin{pmatrix}0\\1\\1\end{pmatrix}=\begin{pmatrix}0\\1\\0\\0\end{pmatrix},\quad \frac{1}{6}\mathbf{A}\begin{pmatrix}2\\-1\\1\end{pmatrix}=\begin{pmatrix}1/\sqrt{2}\\0\\1/\sqrt{2}\\0\end{pmatrix},\quad \frac{1}{3}\mathbf{A}\begin{pmatrix}-1\\-1\\1\end{pmatrix}=\begin{pmatrix}0\\0\\0\\1\end{pmatrix}.$$

Find $\mathbf{U}$. Give exact answer.

(c) (3 points) Use the information given in (b) to find $\mathbf{A}$. Give exact answer.

19. A linear transformation $T:\mathbb{R}^3\to\mathbb{R}^3$ is given by

$$T\left(\begin{pmatrix}x\\y\\z\end{pmatrix}\right)=\begin{pmatrix}x-y-2z\\-2x+2y+2z\\-y-z\end{pmatrix}.$$

(a) (2 points) Write down the standard matrix of the transformation T.

(b) (3 points) Find a nonzero vector $\mathbf{u}$ in $\mathbb{R}^3$ such that $T(\mathbf{u})=\mathbf{u}$. Explain how you derive your answer.

(c) (4 points) Find a vector $\mathbf{u}$ such that $T(\mathbf{u})=\begin{pmatrix}5\\-4\\1\end{pmatrix}$. Explain how you derive your answer.